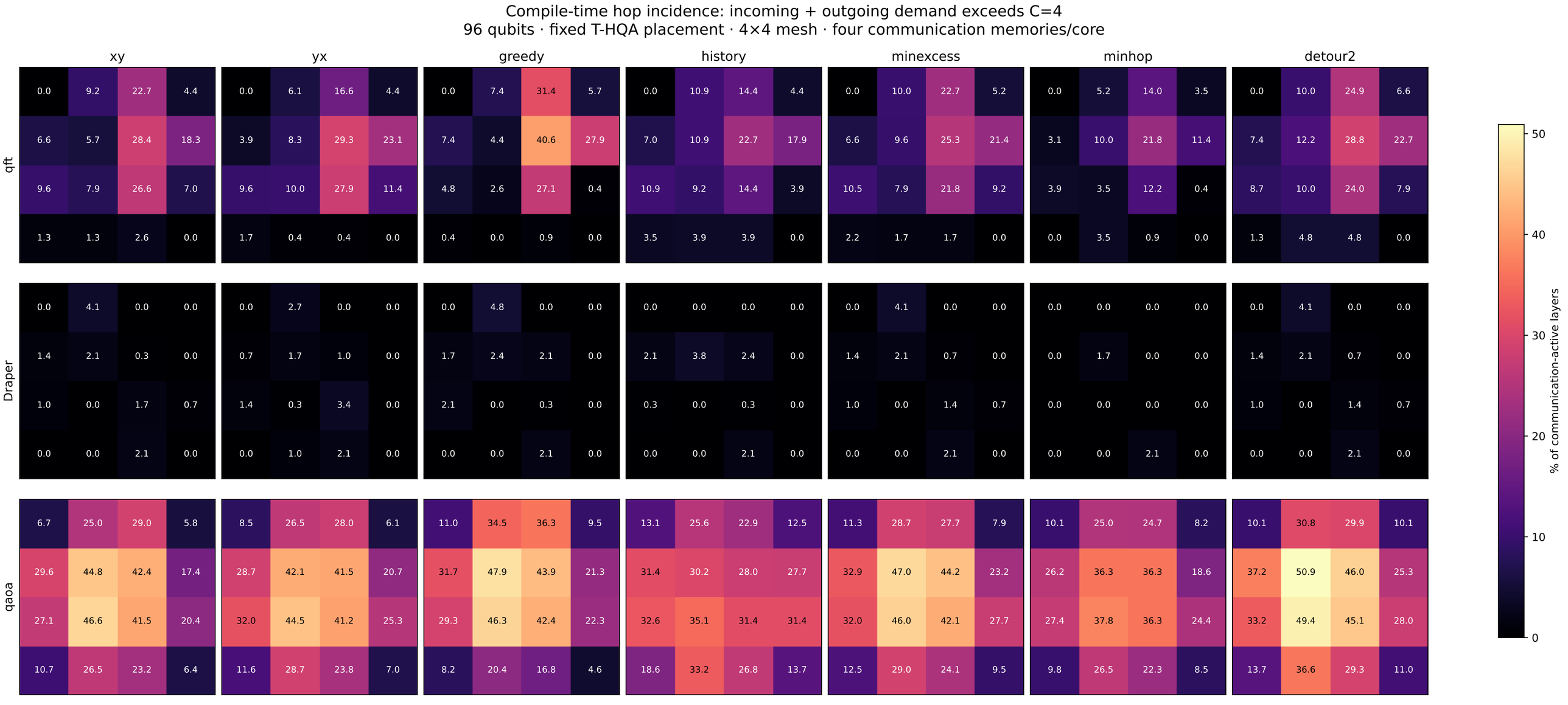
Compile-time hop incidence: incoming + outgoing demand exceeds C=4
96 qubits · fixed T-HQA placement · 4×4 mesh · four communication memories/core
xy
yx
greedy
history
minexcess
minhop
detour2
qft
Draper
qaoa
0.0 9.2 22.7 4.4
6.6 5.7 28.4 18.3
9.6 7.9 26.6 7.0
1.3 1.3 2.6 0.0
0.0 6.1 16.6 4.4
3.9 8.3 29.3 23.1
9.6 10.0 27.9 11.4
1.7 0.4 0.4 0.0
0.0 7.4 31.4 5.7
7.4 4.4 40.6 27.9
4.8 2.6 27.1 0.4
0.4 0.0 0.9 0.0
0.0 10.9 14.4 4.4
7.0 10.9 22.7 17.9
10.9 9.2 14.4 3.9
3.5 3.9 3.9 0.0
0.0 10.0 22.7 5.2
6.6 9.6 25.3 21.4
10.5 7.9 21.8 9.2
2.2 1.7 1.7 0.0
0.0 5.2 14.0 3.5
3.1 10.0 21.8 11.4
3.9 3.5 12.2 0.4
0.0 3.5 0.9 0.0
0.0 10.0 24.9 6.6
7.4 12.2 28.8 22.7
8.7 10.0 24.0 7.9
1.3 4.8 4.8 0.0
0.0 4.1 0.0 0.0
1.4 2.1 0.3 0.0
1.0 0.0 1.7 0.7
0.0 0.0 2.1 0.0
0.0 2.7 0.0 0.0
0.7 1.7 1.0 0.0
1.4 0.3 3.4 0.0
0.0 1.0 2.1 0.0
0.0 4.8 0.0 0.0
1.7 2.4 2.1 0.0
2.1 0.0 0.3 0.0
0.0 0.0 2.1 0.0
0.0 0.0 0.0 0.0
2.1 3.8 2.4 0.0
0.3 0.0 0.3 0.0
0.0 0.0 2.1 0.0
0.0 4.1 0.0 0.0
1.4 2.1 0.7 0.0
1.0 0.0 1.4 0.7
0.0 0.0 2.1 0.0
0.0 0.0 0.0 0.0
0.0 1.7 0.0 0.0
0.0 0.0 0.0 0.0
0.0 0.0 2.1 0.0
0.0 4.1 0.0 0.0
1.4 2.1 0.7 0.0
1.0 0.0 1.4 0.7
0.0 0.0 2.1 0.0
6.7 25.0 29.0 5.8
29.6 44.8 42.4 17.4
27.1 46.6 41.5 20.4
10.7 26.5 23.2 6.4
8.5 26.5 28.0 6.1
28.7 42.1 41.5 20.7
32.0 44.5 41.2 25.3
11.6 28.7 23.8 7.0
11.0 34.5 36.3 9.5
31.7 47.9 43.9 21.3
29.3 46.3 42.4 22.3
8.2 20.4 16.8 4.6
13.1 25.6 22.9 12.5
31.4 30.2 28.0 27.7
32.6 35.1 31.4 31.4
18.6 33.2 26.8 13.7
11.3 28.7 27.7 7.9
32.9 47.0 44.2 23.2
32.0 46.0 42.1 27.7
12.5 29.0 24.1 9.5
10.1 25.0 24.7 8.2
26.2 36.3 36.3 18.6
27.4 37.8 36.3 24.4
9.8 26.5 22.3 8.5
10.1 30.8 29.9 10.1
37.2 50.9 46.0 25.3
33.2 49.4 45.1 28.0
13.7 36.6 29.3 11.0
% of communication-active layers
0
10
20
30
40
50